

MAGNETO-OPTICAL PROPERTIES OF FERROMAGNETIC CHROMIUM TRIHALIDES

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Abstract—Very large rotations of the axis of linear polarization may be observed in visible light passing through ferromagnetic chromium trihalides. The greatest effects are apparently associated with an absorption band edge which, for CrBr_3 , is located in the green. This paper presents the spectral variation of that rotation and of the corresponding absorption band edge. We consider a model consisting of a Cr^{3+} ion in an octahedron of Br^- ions, subjected to a molecular field. It is shown that the absorption band edge in CrBr_3 corresponds to charge transfer transitions in which an electron in a bromine orbital is promoted into an orbital localized on the central chromium ion. The rotation arises as a consequence of the splitting of the excited energy levels by the spin-orbit coupling and the trigonal crystal field.

INTRODUCTION

THE optical properties of the ferromagnetic chromium trihalides enable us to study and to understand many of the important excited energy levels^(1,2) of these crystals. This ability arises from the fact that CrBr_3 and CrCl_3 are moderately transparent over a wide spectral range and there is a good deal of structure in the absorption within this range. In addition, we have the valuable added data of the magneto-optical rotations. Spectral measurements of these rotations provide additional tests of any theoretical interpretation. On an experimental plane, they enable us to distinguish transitions which have a splitting associated with the magnetization in the presence of stronger transitions which are not so split. In Part I of this paper we will describe the magnetic and optical properties of CrCl_3 , and CrBr_3 and CrI_3 with special attention to the large rotations near the absorption band edge. In Part II we will present the theory of these absorptions and rotations.

PART I: PROPERTIES OF THE CHROMIUM TRIHALIDES

1. Crystal structure

A recent paper by MORISIN and NARATH⁽³⁾ has shown that, at low temperatures, the crystal structures of CrCl_3 and CrBr_3 are identical. At about 240°K CrCl_3 undergoes a first order transition to a related structure of lower symmetry with which we shall not concern ourselves. The dominant feature⁽⁴⁾ of the CrBr_3 structure is a covalently bonded "sandwich" in which a hexagonal net of Cr^{3+} ions lies between two close packed layers of halogen ions. While the bonding within the sandwich is predominantly covalent, successive sandwiches are held together only by weak van der Waals bonds. The halogen ions are very nearly h.c.p. Until the Morisin and Narath paper it was thought that the chloride and the iodide had a different structure in which the halogens are cubic close packed. In light of the Morisin and Narath paper, it seems very likely that the iodide also has that same structure. Figure 1⁽⁵⁾ illustrates the bromide-chloride structure. In it, the chromium ion is surrounded by a very nearly regular octahedron of halogen ions. The point

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Table 1

Compound	CrCl ₃	CrBr ₃	CrI ₃
Space group	C_{3i}^2 below $\sim 238^\circ\text{K}^{(a)}$	C_{3i}^2 below $\sim 400^\circ\text{K}$	Probably C_{3i}^2
Halogen packing		h.c.p. for C_{3i}^2	
Cr site point symmetry	C_3	C_3	C_3
Hexagonal cell a_0	5.942 Å ^(c)	6.26 Å ^(b)	6.86 Å
c_0	17.333 Å ^(c)	18.20 Å ^(b)	19.88 Å
X-ray density	2.95 g/cm ³	4.75 g/cm ³	5.36 g/cm ³
T_c	16.8°K ^(d)	32.5°K ^(e)	68°K ^(f)
$4\pi M_0$	3880 (est.)	3390 ^(g)	2690 ^(f)
$2K/M$ at 0°K	~ 0 Oe ^(h)	6500 Oe ⁽ⁱ⁾	28,600 Oe ^(f)
Cr ³⁺ ions/cm ³	1.12×10^{22}	1.0×10^{22}	0.75×10^{22}

(a) NARATH A. and DAVIS H. L., *Phys. Rev.* **137**, A163 (1965).

(b) BRAEKKEN H., *K. Norske Vidensk. Selsk. Forh.* **5**, No. 11 (1932).

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(f) DILLON J. F., JR. and OLSEN C. E., *J. Appl. Phys.* **36**, 1259 (1965).

(g) TSUBOKAWA I., *J. Phys. Soc. Japan* **15**, 1664 (1960).

(h) DILLON J. F., JR. and REMEIKA J. P., *Proceedings of the XIth Colloque Ampere, Eindhoven*, 2-7 July, 1962, pp. 480-487.

(i) DILLON J. F., JR., *J. Appl. Phys.* **33**, Suppl. 1191 (1962).

symmetry for the Cr³⁺ site is C_3 . But it is clear that the leading term in the expansion of the crystal field will be cubic, and that this is followed by a trigonal term. Table 1 gives detailed information on the space group, cell dimensions, and magnetic constants of the three magnetic chromium trihalides.

Chromium tribromide^(5,6) and chromium trioxide⁽⁷⁾ are true ferromagnets. Below T_c a spontaneous magnetization exists which is broken up into domains in the absence of an applied field. Chromium trichloride,⁽⁸⁾ on the other hand is a metamagnet. Below T_c , in the absence of an applied field, the spins are arranged antiferromagnetically. In a relatively small field, a few thousand oersteds, they become ferromagnetically arrayed. Thus, if our experiments are done in a saturating field, CrCl₃, CrBr₃ and CrI₃ are ferromagnets.

The occurrence of interesting magneto-optical properties in these magnetic chromium trihalides suggests that other compounds with similar structures might also display such properties. Extrapolations to such metamagnetic layer structure crystals as FeCl₂, CoCl₂ and NiCl₂ are promising. In addition, we note that BiI₃, SbI₃ and AsI₃ provide non-magnetic isomorphs for CrCl₃ and

CrBr₃. Thus they could be used as hosts for chromium at low concentrations. Such crystals would be valuable for optical and resonance studies of single ions and pairs of ions.

2. Physical properties

Our experimental work on CrBr₃ has been circumscribed by the physical properties of the crystal, particularly its room temperature mechanical properties. Let us describe these qualitatively. The thin sheet samples used in both the optical and microwave experiments bend exceedingly easily, and there is only a small elastic range. The ease with which a permanent deformation is effected presumably represents the ease of slip between the covalently bonded sandwiches mentioned above. In fact, if a relatively thick crystal is subjected to a shear by putting it between two glass slides, the crystal shears like a new deck of cards. The mechanical properties of CrBr₃ are quite similar to those of crystalline graphite.

CrCl₃ is somewhat better mechanically at room temperature where it has a different structure. The yield point for bending is not so low as in the bromide. Altogether the crystal is not as "slippery". However the chloride crystals we have encountered

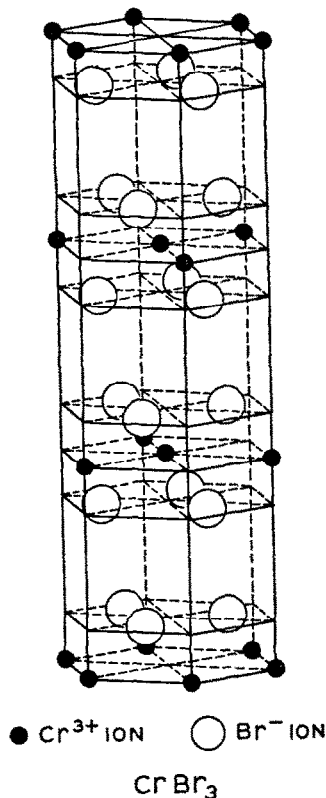


FIG. 1. The structure of CrBr_3 , of CrCl_3 at low temperatures, and probably of CrI_3 . The drawing is made from that given by TSUBOKAWA.⁽⁵⁾ The basic structural element is the halogen-chromium-halogen sandwich.

all seemed to consist of much thinner layers than those of the bromide.

The CrI_3 crystals* were less conspicuously layer-like than CrBr_3 crystals. Some of the material is granular, and some consists of rounded flakes. From the mechanical properties of the three magnetic trihalides it appears that the bonding between layers increases markedly as we go to the heavier halogens.

3. Optical absorption

The optical absorption coefficients of these chromium trihalides show a good deal of structure. Many of the lower lying sharp lines and broad bands typical of the Cr^{3+} ion in a crystal environment with octahedral symmetry are observed.

* CrI_3 crystals were grown by C. E. Olsen, Los Alamos Scientific Laboratories.

The solid lines in Fig. 2 show the absorption coefficient for the three trihalides as measured at 1.5°K . As will be seen below, the positions of the bands and lines are largely accounted for by crystal field theory. Therefore the crystal field assignment for the levels involved has been placed on the drawing next to each peak. In addition to these crystal field transitions the high energy end of each absorption curve is determined by a strong absorption band edge. In the case of CrCl_3 the edge is very close to the limit of the spectral region which we were able to examine, so we cannot say anything about the course of the absorption above this edge.

In the case of the bromide, we know from the recent work of JUNG⁽⁹⁾ that the absorption coefficient in the band arises from two bands. At room temperature these are at $24,310$ and $29,580\text{ cm}^{-1}$. In transmission, even the thinnest samples with which we have been able to work (about 1μ) are opaque from just below $22,000\text{ cm}^{-1}$ to above $25,000\text{ cm}^{-1}$.

The magnetic state of the specimen must be considered in the measurement of optical absorption. For the tribromide, and the iodide, both ferromagnets, it is necessary first that the specimen be magnetically saturated in order to avoid diffraction of light by the domain structure. Absorption coefficients measured on demagnetized samples show spurious structure arising from this diffraction. The topic is discussed in Ref. 10.

Aside from the diffraction and ordinary optical anisotropy we find that the details of the absorption spectrum vary slightly with the crystal orientation of the magnetization. In the case of a sample magnetized along the c -axis which is along the line of sight, the absorption structure is different for the two senses of circular polarization. For the magnetization normal to the c -axis and the line of sight, the absorption must be measured separately for the light polarized parallel and perpendicular to the magnetic field.

When the absorption with $M\parallel c$ is measured for the two senses of circular polarization significant differences are found. The structure of the doublets is distinctly different. More important for our purposes in this paper, the position of the absorption edge is somewhat different for the two senses of circular polarization. It is the associated difference in the dispersion for the two senses which

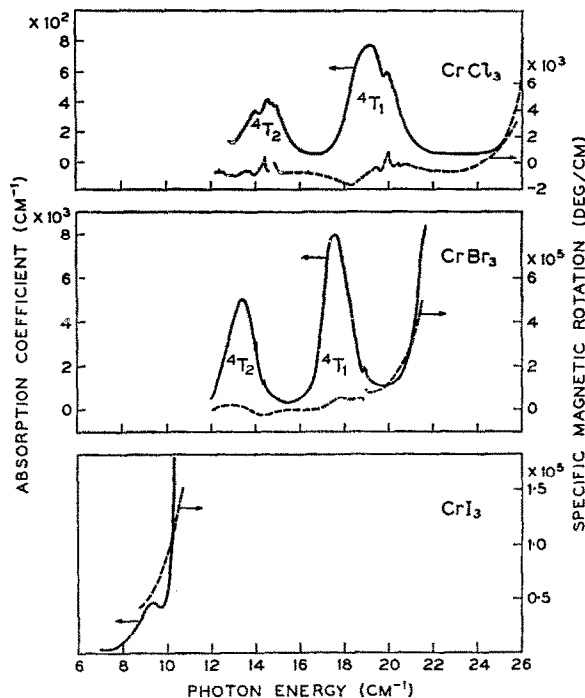


FIG. 2. The optical absorption coefficient $\alpha(I = I_0 e^{-\alpha t})$ (solid curves) and the specific magnetic rotation ρ ($R = t\rho(\lambda)M \cdot l/|M|$) (dashed curves) of CrCl_3 , CrBr_3 , and CrI_3 . Measurements were made at 1.5°K with a saturating field along the line of sight. In the chloride and bromide slightly different absorption curves are measured for the two senses of circular polarization.

accounts for the magnetic rotations to be discussed below. Figure 3 shows schematically for CrBr_3 this difference in absorption coefficient for right and left handed circularly polarized light. It also shows that the 2T_2 line appears only in the sense for which the edge is apparently higher in energy. We measure differences between the position of the edge for the two senses of ~ 0 for CrCl_3 , ~ 200 cm^{-1} for CrBr_3 and ~ 90 cm^{-1} for CrI_3 .

Anisotropic behavior in the strength and position of lines and bands is obviously valuable in constructing and evaluating a theory of the energy levels in these compounds.

4. Magnetic rotation

Just because the real part of the refractive index is different for the two senses of circularly polarized light, linearly polarized light passing through a magnetic crystal undergoes a rotation

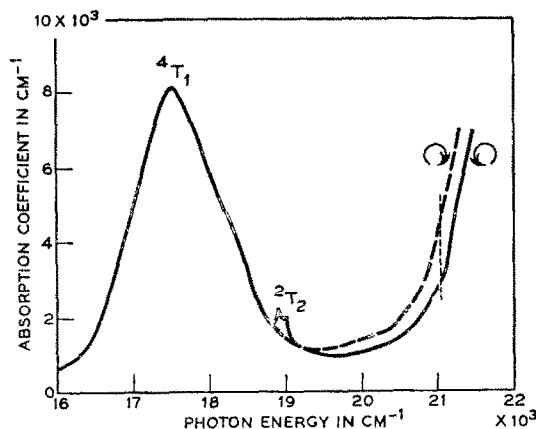


FIG. 3. For axially magnetized CrBr_3 at 1.5°K the absorption band edge is about 200 cm^{-1} higher for one sense of circular polarization than for the other. It is higher for that sense in which the 2T_2 peak occurs.

of its axis of polarization. This magnetic rotation is bound up with the magnetization, and we may describe it by the equation

$$R = t\rho(\lambda)M \cdot l/|M|. \quad (1)$$

In this, R is the rotation observed, t is the thickness of the sample, ρ (a function of wavelength λ) is the rotation per unit thickness observed when the magnetization is parallel to the line of sight. The fraction $M \cdot l/|M|$ represents the fractional projection of the magnetization on the line of sight. Reversing the magnetization reverses the sense of the rotation. If M is perpendicular to l there is no rotation. Aside from determining the direction of the magnetization, moderate magnetic fields do not affect the rotation. Huge fields would presumably have an effect on the rotation.

The rotation data given in Fig. 2 were taken with a saturating field applied perpendicular to the major planes of our thin sheet samples. For the tribromide they show a dispersion-like structure associated with the two quartet absorption peaks and some added variation at the position of the 2T_2 absorption. By far the largest values of specific rotation are measured near the absorption band edge itself. Here we have measured specific rotations of over half a million degrees per centimeter. But there are very large rotations in regions where the absorption is low. In the green at $20,000\text{ cm}^{-1}$, a minimum in the absorption, the specific rotation is $130,000^\circ/\text{cm}$. The trichloride rotation shown in Fig. 2 shows a dispersion at the 4T_1 position. Narrower peaks corresponding to the various doublet transitions are much more prominent here than in the tribromide. The specific rotation rises toward the band edge, but it never seems to approach the values seen in CrBr_3 .

5. Interpretation of the spectrum

The fit of crystal field theory to the spectra of chromium trichloride and tribromide was explored in a paper by WOOD *et al.*⁽¹¹⁾ Table 2 presents an extract from that paper. The crystal field parameters Dq , B and C , and C/B were evaluated according to a specific procedure which gave emphasis to the positions of the lower lying levels. The parameter Dq is a measure of the cubic crystal field; B and C are the parameters for

Table 2. Crystal field parameters of CrCl_3 and CrBr_3 as given by Wood *et al.*⁽¹¹⁾

	CrCl_3	CrBr_3
$Dq, \text{ cm}^{-1}$	1370	1340
$B, \text{ cm}^{-1}$	550	370
$C, \text{ cm}^{-1}$	3400	3700
C/B	6.3	10.0
Cr-ligand dist., Å	2.38	2.54

Spin-orbit coupling parameter taken as $\zeta = 170\text{ cm}^{-1}$

Coulomb interaction between d electron parameters. The substance of Ref. 11 is the presentation of such parameters for a number of crystals, and a discussion of their significance. In the case of CrCl_3 and CrBr_3 the parameters are chosen in such a way as to fit the position of the 4T_1 , 4T_2 , and 2E levels. The 2T_1 level then computed from these parameters corresponds to an experimental peak, but for both compounds the observed 2T_2 peak is depressed very considerably below its theoretical position. This disparity may arise from interaction with the absorption band edge.

PART II. ORIGIN OF THE LARGE MAGNETIC ROTATION OF THE CHROMIUM TRIHALIDES

1. Preliminary remarks

Although attempts to explain the large rotation of the polarized light observed in ferromagnetic metals in terms of microscopic theory were made at the beginning of this century, HULME⁽¹²⁾ first pointed out that the spin-orbit interaction in the excited states played an essential role in producing such large rotation. Based on the Heitler-London model for ferromagnetics, he showed that the difference between the indices of refraction of the right and left hand circularly polarized light arose from the splitting of the excited state by the spin-orbit interaction. After that, ARGYRES⁽¹³⁾ developed a more refined theory, using the band theory.

Recently in considering yttrium iron garnet, CLOGSTON^(14,15) showed that the rotation associated with its absorption band edge was due to the spin-orbit interaction.

The purpose of Part II is to elucidate the origin of the anomalously large magnetic rotation of chromium trihalides, by determining the structure at the absorption band edge. Since these compounds are insulators, it is reasonable to adopt the Heitler-London model⁽¹⁶⁾ for the unpaired spins of the Cr^{3+} ions. And further, for the exchange interaction between them we adopt the molecular field approximation.⁽¹⁷⁾ On these assumptions, first we specify the energy levels of the localized excited states, corresponding to the absorption band edge, in terms of molecular orbitals. Second, we discuss the effect of the spin-orbit interaction on these states and derive the relation between the specific rotation and the energy splitting due to the spin-orbit interaction and the trigonal field. Third, we explore the electron configurations corresponding to the absorption band edge. For these configurations, the spin-orbit coupling constant of the electron transfer state is computed and compared with the experimental results.

2. The MO model for the localized excitation

The overall character of the absorption spectrum below the absorption band edge can be accounted for by considering a cubic crystal field acting on a single Cr^{3+} ion. For dealing with the band edge, our model consists of a single Cr^{3+} ion subjected to a molecular field and surrounded by a nearly octahedral arrangement of six halogen ions. As noted in the section on crystal structure the chromium site is not at an inversion center. Thus our model neglects the lack of inversion symmetry. It will be shown that this simple model can account for the optical characteristics of the edge. Since the absorption band edge has a much stronger intensity than that of the absorption spectrum below the edge, the transitions corresponding to the edge must be allowed ones. This means the parity of the states before and after transitions must be different, because we assume O_h site symmetry for a Cr^{3+} ion surrounded by six neighboring halogen ions. For such transitions there are two main assignments: (1) intra-atomic type transitions from a $3d$ state to a $4p$ state of the Cr^{3+} ion; and (2) electron transfer transitions in which an electron from a halogen ion is removed to the Cr^{3+} ion. Since we assume that the excitation is localized in the region of a particular Cr^{3+}

ion and its six surrounding halogen ions, it is convenient to consider the assignment of the absorption band edge in terms of the molecular orbitals in which $3d$, $4s$ and $4p$ orbitals of the Cr^{3+} ion are combined with linear combinations of p orbitals of the halogen ions. Such molecular orbitals are classified into σ orbitals of T_{1u} , E_g and A_{1g} symmetry and π orbitals of T_{2g} symmetry. Besides these bonding and antibonding orbitals, linear combinations of p_π wave functions of the six halogen ions lead to orbitals of T_{1u} , T_{2u} and T_{1g} symmetry which do not combine with the orbitals of the Cr^{3+} ion (we call them nonbonding π orbitals).^{*} Expressions for some of these orbitals are given in Table 1 of Appendix III. Their schematic energy diagram is shown in Fig. 4. The ordering of the orbital energy has been determined by considering the nodal pattern of each molecular orbital.⁽¹⁸⁾ In the present case all orbitals up to the t_{1g} nonbonding state are completely occupied, and further, the $t_{2g}(\pi)$ antibonding orbital is occupied by three electrons which have parallel spins. Thus we can make the following three assignments for the absorption band edge, the transition of an electron (1) from a t_{2g} antibonding to a t_{1u} antibonding orbital (an "internal" process); (2) from a t_{1u} bonding to a t_{2g} or e_g antibonding orbital ("σ process"); (3) from a t_{1u} or t_{2u} nonbonding to t_{2g} or e_g orbital ("π process"). The first one corresponds to the intra-atomic transition and the last two to the so-called electron transfer transitions. These assignments are indicated in Fig. 4.

No matter what assignments may be chosen among these three, it should be noticed that only the states of ${}^4T_{2u}$ symmetry are accessible from the ${}^4A_{2g}$ ground state by electric dipole transitions. We will show later which assignment actually corresponds to the absorption band edge.

3. Effect of spin-orbit interaction on the 4T_2 state and the magnetic rotation

In this section we will consider the effect of the spin-orbit interaction on the 4T_2 state and will show that this leads to the magnetic rotation of the chromium trihalides. The T_2 state is orbitally

^{*} Because of their small overlap, we neglect the combination between $4p$ orbitals of a Cr^{3+} ion and halogen $p\pi$ orbitals of T_{1u} symmetry.

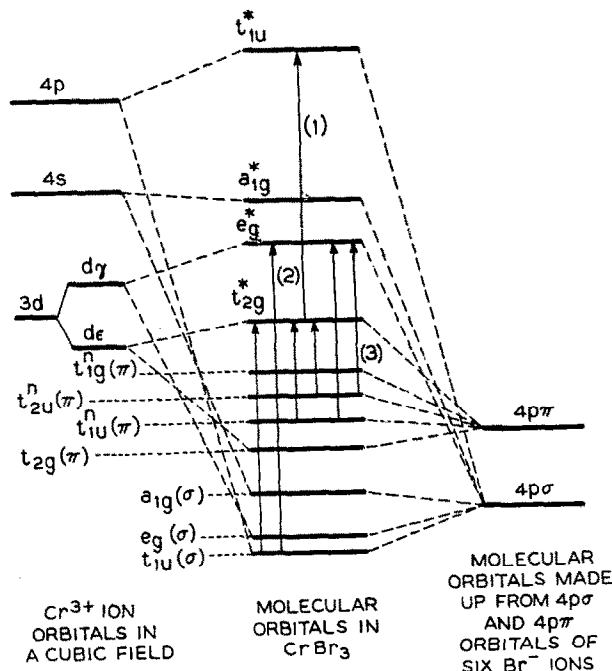


FIG. 4. Schematic energy diagram of CrBr_3 . All orbitals up to $t_{1g}^n(\pi)$ which have mainly the character of Br^- orbitals are completely occupied. The orbitals having the character of the $\text{Br}^- \sigma$ and π orbitals are indicated by (σ) and (π) . The orbitals with an asterisk are all antibonding ones, and they have mainly metallic character. The t_{2g}^* orbitals are occupied by three electrons, and the remaining anti-bonding orbitals are vacant.

triply degenerate. Therefore, the spin-orbit interaction acts to lift the degeneracy of 4T_2 state. Since the Cr^{3+} ions reside in trigonally distorted sites, they are subjected to an electrostatic field with trigonal components in addition to the large cubic field. Thus we will consider the combined effect of the spin-orbit interaction and the trigonal field on the 4T_2 state.

Let us solve the effective Hamiltonian for the 4T_2 state. Using the orbital angular momentum operator $L(L+1) = 2$, this may be written

$$H_{\text{eff}} = \Delta(1 - L_z^2) - \lambda(\alpha^4 T_2) L \cdot S + g\mu_B S_z H_{\text{ex}} \quad (2)$$

where Δ indicates the trigonal splitting of the T_2 state, $\lambda(\alpha^4 T_2)$ is the spin-orbit coupling constant of the state 4T_2 with a particular electron configuration α and the last term represents the exchange interaction between the Cr spins, as was mentioned before, for the exchange interaction we

have adopted the molecular field approximation. The z -axis is taken along the direction of the molecular field H_{ex} which is experimentally directed along the c -axis by the applied field. In solving (2), we adopt a representation in which L_z and S_z are diagonal.

Assuming $|\lambda|$ and $|\Delta| \gg \mu_B H_{\text{ex}}$, we will solve first the Hamiltonian (2) in the absence of the exchange field. In this case, the 6×6 matrix for H_{eff} breaks up into

$$\begin{bmatrix} +\frac{3}{2}\lambda & -\sqrt{(\frac{3}{2})}\lambda & 0 \\ -\sqrt{(\frac{3}{2})}\lambda & \Delta & -\sqrt{(2)}\lambda \\ 0 & -\sqrt{(2)}\lambda & -\frac{1}{2}\lambda \end{bmatrix} \quad (m = \pm \frac{1}{2})$$

$$\begin{bmatrix} \Delta & -\sqrt{(\frac{3}{2})}\lambda \\ -\sqrt{(\frac{3}{2})}\lambda & -\frac{1}{2}\lambda \end{bmatrix} \quad (m = \pm \frac{3}{2})$$

$$(-\frac{3}{2}\lambda) \quad (m = \pm \frac{5}{2})$$

Since the Hamiltonian commutes with $L_z + S_z$, each Kramers doublet is specified by $m = \pm 1/2, \pm 3/2, \pm 5/2$ where m is the eigenvalue of the operator $L_z + S_z$. The eigenfunctions corresponding to the states $m_z = \pm 1/2, \pm 3/2$ and $\pm 5/2$ are

$$|m = \pm \frac{1}{2}\rangle = a|L_z = \mp 1, S_z = \pm \frac{3}{2}\rangle + b|L_z = 0, S_z = \pm \frac{1}{2}\rangle + c|L_z = \pm 1, S_z = \mp \frac{1}{2}\rangle \quad (3a)$$

$$|m = \pm \frac{3}{2}\rangle = \alpha|L_z = 0, S_z = \pm \frac{3}{2}\rangle + \beta|L_z = \pm 1, S_z = \pm \frac{1}{2}\rangle \quad (3b)$$

and

$$|m = \pm \frac{5}{2}\rangle = |L_z = \pm 1, S_z = \pm \frac{3}{2}\rangle. \quad (3c)$$

The energy variation of the six Kramers doublets as a function of $\Delta/|\lambda|$ is shown in Fig. 5.

In the presence of the exchange field, each

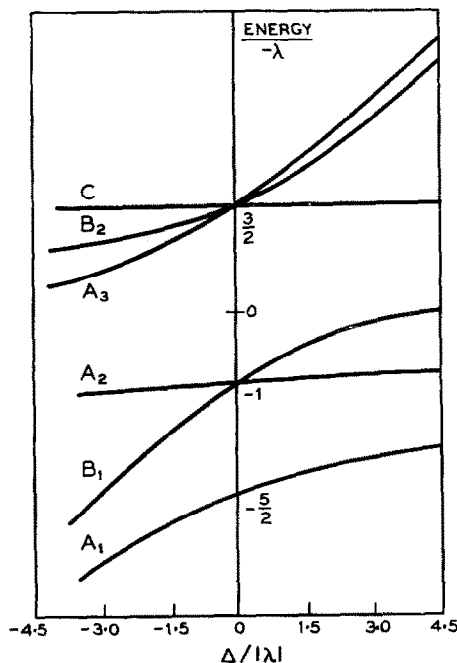


FIG. 5. The energy level curves of the $4T_2$ state in the spin-orbit interaction and trigonal field. The positions of the energy levels on the ordinates give the splitting of the $4T_2$ state by the spin-orbit interaction.

doublet is split, the ground state $4A_2$ is also split. This splitting is given by $Jz\bar{S}$, where we adopted the form $J\Sigma_i \mathbf{S}_i \cdot \mathbf{S}_i$ for the original ferromagnetic nearest neighbor interactions. The quantity z is the number of nearest neighbors and $\bar{S}$ is the average value of a single Cr spin along the magnetization. In the case of CrBr_3 , if we take only the interactions between the Cr spins in the same layer into account, $Jz\bar{S}$ is 17 cm^{-1} at the zero temperature, where we used 3.8 cm^{-1} determined by GOSSARD *et al.*⁽¹⁹⁾ for J the exchange integral and $z = 3$. Thus at 1.5°K , the temperature of the measurement, only the lowest component of the ground state $S_z = 3/2$ will be populated. The quantity S_z does not change in an electric dipole transition. Thus, transitions from the $S_z = 3/2$ component of the ground state to the $m = 1/2$ and $5/2$ components of the upper state are allowed for right (E_+) and left (E_-) circular polarizations respectively, where $E_\pm = E_x \pm iE_y$, E_x and E_y being x and y components of the external electric field. The intensity of the transition to $m = 1/2$ is proportional to a^2 , which deviates from unity by means of the spin-orbit interaction. As can be seen in the table of Appendix I, a^2 is large for the state A_1 when $\Delta/|\lambda|$ is positive and large and for the state A_2 when $\Delta/|\lambda|$ is negative and large. In the case where a^2 is negligibly small for the state A_3 , we may consider that the energy difference A_1 and C (A_2 and C) gives the difference between the absorption band edges for two senses of circular polarization in the case of positive (negative) trigonal field. As seen in Fig. 5, the energy difference between A_1 and C (A_2 and C) is determined by the value of λ , which depends critically on the configuration α of the excited state, as will be seen later.

The specific rotation $\rho(\omega)$ in equation (1) is defined as

$$\rho(\omega) = \frac{\omega}{2c} \text{Re}(n_+ - n_-), \quad (4)$$

where n_+ and n_- are the complex refractive indices for the right and left circular polarizations, respectively, and Re means the real part of the expression. As will be derived in Appendix II, the difference $n_+ - n_-$ is related to the xy component of the polarizability tensor α_{xy} . The quantity

$(n_+ - n_-)$ is given by

$$n_+ - n_- = \frac{4\pi N}{n\hbar} \times \sum_{\alpha} p_{\alpha}^2 (\omega - i\Gamma_{\alpha}) \left\{ \frac{1}{(\omega_{C(\alpha)g}^2 - \omega^2 + \Gamma_{\alpha}^2) + 2i\omega\Gamma_{\alpha}} - \sum_{=1,2,3} \frac{a_{A_i}^2(\alpha)}{(\omega_{A_i(\alpha)g}^2 - \omega^2 + \Gamma_{\alpha}^2) + 2i\omega\Gamma_{\alpha}} \right\} \quad (5)$$

where $n = 1/2(n_+ + n_-)$, p_{α}^2 is the average dipole moment for the transition from the ground state 4A_2 to the excited state 4T_2 with a particular electron configuration α , $\hbar\omega_{A_i(\alpha)g} = \epsilon_{A_i(\alpha)} - \epsilon_g$, $\hbar\omega_{C(\alpha)g} = \epsilon_{C(\alpha)} - \epsilon_g$, N is the number of chromium ions/unit vol. and Γ_{α} is a damping factor. The quantities $\epsilon_{A_i(\alpha)}$, $\epsilon_{C(\alpha)}$ and ϵ_g are the energies of the sub-states A_i and C of the 4T_2 excited state with a particular configuration α and of the ground state 4A_2 . Σ_{α} is the summation taken over the electron configuration from which the 4T_2 state arises.

Equation (5) is valid only at very low temperature in which the $S_z = \frac{3}{2}$ component is overwhelmingly populated. If we do not take the spin-orbit interaction into account in the 4T_2 state, $\epsilon_C = \epsilon_{A_2}$, $a_{A_2}^2 = 1$ and $a_{A_1}^2 = a_{A_3}^2 = 0$ (or $\epsilon_C = \epsilon_{A_1}$, $a_{A_1}^2 = 1$ and $a_{A_2}^2 = a_{A_3}^2 = 0$) in the case of negative (or positive) trigonal field. Thus the rotation vanishes. Therefore, we can conclude that the spin-orbit interaction in the 4T_2 state plays an essential role in producing the magnetic rotation in CrBr_3 . Note particularly that in the low temperature region of this analysis and of our experiments the specific rotation does not depend on the applied magnetic field or even on the exchange field. This is the characteristic in which the rotations discussed here differ from the usual Faraday effect in which the level splitting is determined by an applied field. Obviously, just below the Curie point (whose value reflects the exchange field), the specific rotation does depend on the magnetization and thus on the applied field.

4. Assignment of an electron configuration to the absorption band edge

In this section we will determine electron configurations corresponding to the absorption band edge from among the three possible assignments given in Section 2. In Table 3 we tabulate

Table 3. The number of the states having T_2 symmetry in each assignment

	Assignment	The number
(1)	$t_{2g}^* \rightarrow t_{1u}^*$	1
(2a)	$t_{1u} \rightarrow t_{2g}^*$	1
(2b)	$t_{1u} \rightarrow e_g^*$	1
(3a)	$t_{1u}^n \rightarrow t_{2g}^*$	1
(3b)	$t_{1u}^n \rightarrow e_g^*$	1
(3c)	$t_{2u}^n \rightarrow t_{2g}^*$	1
(3d)	$t_{2u}^n \rightarrow e_g^*$	1

the number of the states having T_2 symmetry in the three types of assignment. Such a number can easily be calculated by reducing the product of the irreducible representations into the irreducible representations of the O_h group. For example, in the assignment (1), the final states of the transition are constructed by coupling a t_2 hole to a t_1 electron. Since $T_1 \times T_2 = A_2 + E + T_1 + T_2$, the number of the T_2 states in this assignment is one.

Thus the strong absorption band consists of transitions to these T_2 states. Energetically, we expect the states arising from the transitions (3) to appear at the lower frequency side of the absorption band. Using the parameters given in Table 2, we can show that among the transitions (3), the state arising from the transition t_{1u}^n (or t_{2u}^n) $\rightarrow e_g^*$ has a lower energy than that arising from the transition to t_{2g}^* . Namely, although the former has a higher orbital energy by $10Dq$ than the latter, the intra-atomic exchange energy in the former overcomes the above gain in orbital energy to lower the total energy by $6B + 5C - 10Dq$ below that of the latter state. Thus the transitions (3b) or (3d) occur at lower frequencies than (3a) or (3c).

In the case of the electron transfer transitions, the average dipole moment p is approximately proportional to the product of the lattice constant and the overlap between the d orbital of the metal and the p orbital of the halogen.

Roughly speaking $\int |\phi_{d(e_g)}| |\chi_{p\sigma}| d\tau \sim 0.1$ and $\int |\phi_{d(e_g)}| |\chi_{p\pi}| d\tau \sim 0.01$, where $\phi_{d(e_g)}$, $\chi_{p\sigma}$ and $\chi_{p\pi}$ are the chromium $3d(e_g)$ and the halogen p_{σ} and p_{π} orbitals, respectively. Thus the intensities of the transition (2b) are 10 or 10^2 times smaller than those of the intra-atomic transition (1), while

the intensities of the transition (3*b*) or (3*d*) are 10^3 or 10^4 times smaller than those of the transition (1). Therefore, the intensities of the transition (3*b*) or (3*d*) are nearly equal to those of the absorption bands below the edge, and thus, could be observed if not masked by stronger absorption. The intensities of the transitions (2*a*), (3*a*) and (3*c*) are weaker than those of (2*b*), (3*b*) and (3*d*), respectively. We conclude that the lowest lying electron transfer state corresponds to the configuration $t_{2g}^3 e_g^* t_{1u(\text{and } 2u)}^{n5}$. About six thousand wave numbers above this there are states belonging to the configuration $t_{2g}^4 t_{1u(\text{and } 2u)}^{n5}$. Several thousand wave numbers above these there will presumably be the states of the configurations $t_{2g}^3 e_g^* t_{1u}^5$ and $t_{2g}^4 t_{1u}^5$. The intensity of the transition to this may well be 100 times greater than that of the transition between the ground state and either of the other two configurations.

However, this conclusion has been drawn without considering the line shape of the peak corresponding to each transition. In fact, there is a possibility that a stronger peak at higher frequency might mask a weaker peak at lower frequency. Whether or not such a possibility occurs depends on the relative magnitudes of the line width of each peak and of the energy difference between the peak positions. Therefore, from the above discussion only on peak positions we can not definitely say which transition corresponds to the actual absorption band edge. Thus the magnetic rotation at the edge plays an important role as one of valuable extra evidence in finding an electron configuration belonging to the edge.

5. Comparison with experiments and discussions

Rather than calculate the magnetic rotation at the absorption band edge, we will evaluate energy difference between the absorption of two senses of circular polarization.

As we have seen in Fig. 5 with the help of Appendix I, this difference depends critically on a value of the ratio $\Delta/|\lambda|$. Namely, in a region of small values of this ratio where a_3^2 is not so small compared with a_2^2 or a_1^2 , the energy difference for two senses of circular polarization is given by that between *C* and *A*₃, while in a region of large values of the ratio where a_3^2 is negligibly small, the energy difference between *A*₁ and *C* (*A*₂ and *C*) gives that for two senses of circular polarization

in the case of positive (negative) trigonal field. Thus, first of all we have to know values of λ and Δ .

In Appendix III we derive expressions for λ for each electron configuration. The results for the assignments (1), (2*b*) and (3*b*) or (3*d*) are shown in Table 4, where ζ_x , ζ_{3d} and ζ_{4p} are spin-orbit coupling constants of the single *p* electron in the halogen atom *X*, of the single 3*d* and 4*p* electrons in the chromium ion, respectively. S_x is the overlap integral between the *p*_σ and *p*_π orbitals on the neighboring halogen ions.

As seen in this table, the spin-orbit coupling constants for the electron transfer states are expressed in terms of that of the single *p* electron in the halogen atom. This is easily seen from the fact that the orbital angular momentum in the state 4T_2 with the configuration

$$t_{2g}^3 ({}^4A_{2g}) e_g^* ({}^2E_g) ({}^5E_g) t_{1u(\text{or } 2u)}^{n5} ({}^2T_{1u(\text{or } 2u)})$$

or

$$t_{2g}^3 ({}^4A_{2g}) e_g^* ({}^2E_g) ({}^5E_g) t_{1u}^5 ({}^2T_{1u})$$

must arise within the $t_{1u(\text{or } 2u)}^n$ or t_{1u} hole orbital and thus the spin-orbit interaction arises from the iteration of the *p* hole among the halogen ions.

According to the spectroscopy for the free chromium ion, Cl, Br and I atoms, ζ_{3d} , ζ_{4p} , ζ_{Cl} , ζ_{Br} and ζ_I are 290 cm⁻¹, 50 cm⁻¹, 590 cm⁻¹, 2460 cm⁻¹ and 5070 cm⁻¹, respectively.⁽²⁰⁾ Using these values, the spin-orbit coupling constants for the assignments (1), (2*b*) and (3*b*) or (3*d*) are calculated for CrCl₃, CrBr₃ and CrI₃ and their numerical results are also shown in Table 4.

On the other hand, we have no definite knowledge of the sign or the magnitude of Δ . However, we can estimate the order of magnitude for Δ from the following consideration of the anisotropy energy.

Take the anisotropy energy to be $K_1 \cos^2 \theta$ where θ is the angle between the magnetization and the *c*-axis. Following WOLF,⁽²¹⁾ we see that K_1 is composed of two parts, one arising from the dipole interaction, and the other from the crystal field splitting of the Cr³⁺ ion ground state. We write

$$K_1 = NDp(y) + K_{\text{dip.}}$$

In this K_{dip} is the dipole contribution, *D* is the coefficient of S_z^2 in the usual spin Hamiltonian

Table 4. The spin-orbit coupling constants for various electron configurations

Assignment	Final electron configuration	S.O. constant	Numerical values (cm ⁻¹)		
			CrCl ₃	CrBr ₃	CrI ₃
(1)	$t_{2g}^{*2}t_{1u}^*$	$\frac{1}{2}(\zeta_{4p} - \zeta_{3d})$	-120	-120	-120
(2b)	$t_{2g}^{*3}e_g^*t_{1u}^5$	$-\frac{2}{5}Sx\zeta_x$	$\sim +5$	$\sim +30$	$\sim +60$
(3b) or (3d)	$t_{2g}^{*3}e_g^*t_{1u}(\text{or } 2u)^{n5}$	$-\frac{1}{5}\zeta_x$	-120	-480	-1010

and $y = \exp(-g\mu_B H_{ex})$. The function $p(y)$ is a polynomial in y , whose value has been computed as a function of magnetization by Wolf. GOSSARD *et al.*⁽²²⁾ have calculated K_{dip} to be $(2420 \text{ Oe}) \times M/2 = 3.3 \times 10^5 \text{ erg/cm}^3$. On the other hand DILLON⁽²³⁾ observed K_1 for CrBr₃ to be $-9.3 \times 10^5 \text{ ergs/cm}^3$. Since $p(y)$ at the lowest temperature is $3/2$ and $N = 0.970 \times 10^{22}/\text{cm}^3$, ND is $-8.4 \times 10^5 \text{ ergs/cm}^3$ and D is 0.43 cm^{-1} .

In the case of the electron transfer transitions, Δ is related to the trigonal splitting of the single t_{1u} or t_{1u}^n electron which has halogen character. Therefore, it is very difficult to calculate Δ in these cases. However, we can estimate the order of magnitude of Δ from the following conjecture.

In ZnO, DIETZ *et al.*⁽²⁴⁾ recently determined the trigonal splitting of the single d electron, Δ_d , from the analyses of Zeeman spectrum and electron spin resonance for Cu²⁺ in ZnO. On the other hand, from the analysis of exciton spectra, THOMAS⁽²⁵⁾ determined the trigonal splitting of the valence band of ZnO, Δ_p , which is related to the trigonal splitting of the single p electron of oxygen. According to their results, $\Delta_d = -170 \text{ cm}^{-1}$ and $\Delta_p = -330 \text{ cm}^{-1}$. Though we have no information on the correlation between Δ_d and Δ_p for crystals other than ZnO, we are led to assume that Δ_p has the same order of magnitude as Δ_d in ionic crystals such as chromium trihalides.

By means of second order perturbation theory, we can estimate Δ_d from D for chromium trihalides, as follows:

$$\Delta_d = \frac{(10 Dq)^2}{4\lambda^2} D.$$

Since, in CrBr₃, $10 Dq = 13400 \text{ cm}^{-1}$ and $\lambda = 90 \text{ cm}^{-1}$, Δ_d is -2300 cm^{-1} . Of course, we do not

take this value of Δ_d too seriously, since second order perturbation theory sometimes gives unreliable results.

By the same procedure, we can also estimate Δ_d for CrCl₃. As for CrI₃, a value of $10 Dq$ has not been determined. Since the absorption band edge appears at 10000 cm^{-1} in CrI₃, we expect a value of $10 Dq$ less than 10000 cm^{-1} . Assuming 7000 cm^{-1} for $10 Dq$ in CrI₃, we can estimate Δ_d for CrI₃. The values of D semiempirically determined from the anisotropy and the estimated values of Δ_d for CrCl₃, CrBr₃ and CrI₃ are shown in Table 5, together with the observed values of K , where the values of K_{dip} have been taken to be the same as those of CrBr₃. The other constants are derived from Table 1.

Table 5. The values of Δ , D and K for chromium trihalides

	CrCl ₃	CrBr ₃	CrI ₃
$K(\text{ergs/cm}^3)$	0	-9.3×10^5	-3.1×10^6
$D(\text{cm}^{-1})$	-0.10	-0.43	-1.54
$\Delta(\text{cm}^{-1})$	-580	-2400	-2300

Based on the values of λ and Δ obtained above, we will proceed to determine an electron configuration corresponding to the absorption band edge. Take CrBr₃ as an example. From Tables 4 and 5, the values of the ratio $\Delta/|\lambda|$ are about -20, -80 and -5 for the assignments (1), (2b) and (3b) or (3d), respectively. For these values of $\Delta/|\lambda|$, a_3^2 is negligibly small compared with a_2^2 . Thus the difference between the absorption band edges of two senses of circular polarization is given by the energy difference between A_2 and C

for these values of $\Delta/|\lambda|$. They are 360 cm^{-1} , 100 cm^{-1} and 600 cm^{-1} for the assignments (1), (2*b*) and (3*b*) or (3*d*) respectively. Similarly, for CrCl_3 and CrI_3 we can calculate the energy difference between the absorption band edges of two senses of circular polarization for each assignment. The calculated results are also shown in Table 6, together with the observed values for CrCl_3 , CrBr_3 and CrI_3 .

Table 6. The calculated and observed energy differences between the absorption band edges of two senses of circular polarization (cm^{-1})

Assignments	CrCl_3	CrBr_3	CrI_3
(1)	320	360	~ 360
(2 <i>b</i>)	20	~ 100	$\sim 150 \sim 180$
(3 <i>b</i>) or (3 <i>d</i>)	320	600	~ 2500
Observed	small	200	90

Though the calculated results for CrI_3 are not definite, the results for the assignment (2*b*) explain the observed variation of the energy splittings in transmission measurements among the chromium trihalides as well as their magnitudes. Thus we can conclude that in chromium trihalides the absorption band edges observed in transmission measurements correspond to the electron configuration $t_{2g}^*{}^3(^4A_{2g})e_g^*t_{1u}^5$ (the assignment (2*b*)). However, this does not imply that the magnetic rotation near the band edge arises from the transition (2*b*). Examination of equations (4) and (5) shows that the rotation depends on the energy splitting as well as on the oscillator strength of the transition. Thus, given a large splitting, it is quite possible for a large magnetic rotation to arise from a transition so weak as to be difficult to detect in the presence of a strong transition with little or no associated splitting.

As seen in Table 4, the energy differences between the absorption band edges of two senses of circular polarization for the assignment (3*b*) or (3*d*) is about ten times larger than those for the assignment (2*b*). Therefore, even though the oscillator strength for the transition (3*b*) or (3*d*) is ten times smaller than that for (2*b*), the rotations arising from the transition (3*b*) or (3*d*) are of the same order of magnitude as those from (2*b*),

and further, it is expected that these transitions will occur at lower energies. We thus conclude that the magnetic rotation just below the absorption band edge in the chromium trihalides corresponds to the electron transfer state with the configuration $t_{2g}^*{}^3(^4A_{2g})e_g^*t_{1u}^{n5}$ or t_{2u}^{n5} in which an electron is removed from the nonbonding t_{1u}^n or t_{2u}^n orbital to the antibonding e_g^* (the assignment (3*b*) or (3*d*)).

Another set of experimental results supports this conclusion for the case of CrBr_3 . JUNG⁽⁹⁾ has recently succeeded in observing the structure of the electron transfer states in the absorption band edge by measurements of the Kerr rotation and of the optical constants. Ellipsometer measurements show the absorption bands at about 24500 and 29500 cm^{-1} which he attributes to the transitions (2*b*) and (1) respectively. In the measurements of the Kerr rotation he observed a maximum rotation with a positive sign at 23500 cm^{-1} , a relatively sharp negative peak at 26700 cm^{-1} and, further a reversal in the sign of the rotation beyond 27500 cm^{-1} .

Since the Kerr rotation is determined by the imaginary part of $n_+ - n_-$ ⁽⁹⁾ the sign of the rotation is due to that of the spin-orbit coupling constant for each assignment and its magnitude depends on the energy splitting due to the spin-orbit coupling as well as the oscillator strength for each transition, as in the case of the magnetic rotation described above. Looking at the results of CrBr_3 for each assignment in Tables 4 and 6, we can see that the above-mentioned observed variation in the magnitudes and the sign of the Kerr rotation as a function of the wave number is consistent with our results if we assume that the transitions (3*b*) or (3*d*), (2*b*) and (1) correspond to the peaks at 23500, 26700 cm^{-1} and higher energy (29500 cm^{-1} in the reflection measurement) respectively. Thus we can conclude that both the magnetic and Kerr rotations near the absorption band edge arise from the transition (3*b*) or (3*d*).

6. Conclusion

In the experimental part of this paper we have shown the very large specific magnetic rotations associated with the absorption band edges of CrBr_3 and CrCl_3 . A general expression for the complex magnetic rotation has been derived,

based on the Heitler-London model for the unpaired spins of the chromium ions. The energy level splittings which give rise to the large rotations are not due to the applied field, they are not due even to the exchange field. They do, in fact, arise from the spin-orbit interaction of the final state. We have calculated the spin-orbit coupling constants of the final state for several electron configurations. We have estimated the magnitude of the trigonal splitting of the final state from the anisotropy energy and this has enabled us to calculate in detail the difference between absorption band edges for two senses of circular polarization. Comparing theoretical results with the observed difference, we attribute the absorption band edge itself to the electron transfer transition from a $t_{1u}(\sigma)$ orbital to an e_g^* orbital. However, the magnetic rotation at the foot of the edge would seem to arise from the splitting of a much weaker electron transfer transition from a $t_{1u}^n(\pi)$ or $t_{2u}^n(\pi)$ orbital to an e_g^* orbital. The large splitting of this transition is due to the spin-orbit interaction which arises from the iteration of the p_π hole among the halogen ions.

In the present paper we have considered the rotations at temperatures low enough so that only the lowest ground state sublevel is significantly populated. The actual spectrum of magnetic rotations will depend on the width of the initial and final levels and on the thermal population of the ground state sublevels as well as on the spin-orbit interaction of the final state.

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APPENDIX I EIGENFUNCTIONS OF $m = \frac{1}{2}$ STATES

In Table A1 below we give values for the coefficients a , b and c of the eigenfunctions of the $m = \frac{1}{2}$ states for various values of $\Delta/|\lambda|$. This refers back to equation (3a). Among the three states A_1 , A_2 , A_3 that designated A_1 has the lowest energy, and A_3 , the highest.

Table A1. Eigenfunctions of $m = \frac{1}{2}$ states

	A_1			A_2			A_3		
$\Delta/ \lambda $	a_1	b_1	c_1	a_2	b_2	c_2	a_3	b_3	c_3
-6.0	0.324	-0.901	0.288	-0.894	-0.304	0.400	-0.292	-0.389	-0.872
-4.5	0.399	-0.855	0.332	-0.863	-0.221	0.454	-0.319	-0.472	-0.822
-3.0	0.492	-0.781	0.375	-0.789	-0.244	0.551	-0.337	-0.562	-0.753
-1.5	0.597	-0.694	0.403	-0.723	-0.256	0.641	-0.340	-0.680	-0.653
0	0.705	-0.578	0.405	-0.619	-0.258	0.742	-0.318	-0.775	-0.550
1.5	0.802	-0.461	0.382	-0.575	-0.253	0.777	-0.280	-0.847	-0.449
3.0	0.857	-0.377	0.347	-0.442	-0.228	0.868	-0.250	-0.893	-0.366
4.5	0.910	-0.290	0.293	-0.412	-0.219	0.884	-0.213	-0.926	-0.306
6.0	0.921	-0.271	0.282	-0.320	-0.188	0.949	-0.190	-0.952	-0.257

APPENDIX II THE RELATION OF ROTATION TO LEVEL SPLITTING

In this Appendix we will derive a relation between the specific magnetic rotation $\rho(\lambda)$ defined by (1) and the energy splitting due to the spin-orbit interaction. First, we assume the wave to be monochromatic with frequency ω , and to propagate along the c axis (z axis):

$$\mathbf{E} = \mathbf{E}_0 e^{i\omega(t-nz/c)}, \quad (\text{A1})$$

$$\mathbf{H} = \mathbf{H}_0 e^{i\omega(t-nz/c)}, \quad (\text{A2})$$

where $\mathbf{E}_0$ and $\mathbf{H}_0$ are constant vectors and n is the index of refraction.

Substituting (A1) and (A2) into the Maxwell equations

$$\text{rot } \mathbf{E} = -\frac{1}{c} \frac{\delta \mathbf{H}}{\delta t} \quad (\text{A3})$$

$$\text{rot } \mathbf{H} = \frac{1}{c} \frac{\delta \mathbf{E}}{\delta t} + \frac{4\pi}{c} \mathbf{J}, \quad (\text{A4})$$

we obtain

$$n(\mathbf{E} \times \mathbf{k}) = \mathbf{H}, \quad (\text{A5})$$

$$n(\mathbf{H} \times \mathbf{k}) = \mathbf{E} + \frac{4\pi}{i\omega} \mathbf{J}, \quad (\text{A6})$$

where $\mathbf{J}$ is the average current density and $\mathbf{k}$ is the unit vector along the z axis. In equation (A3), the change of the average magnetization with respect to time is assumed to be zero.

In the present case the current $\mathbf{J}$ is the polarization current which arises from the time variation of the dipole moment $\mathbf{P}$ induced by the optical

transition.

$$\mathbf{J} = \frac{\delta \mathbf{P}}{\delta t} \quad (\text{A7})$$

Using the polarizability tensor α , the induced dipole moment $\mathbf{P}$ is expressed by

$$\mathbf{P} = N\alpha \cdot \mathbf{E}_{\text{loc}}, \quad (\text{A8})$$

where $\mathbf{E}_{\text{loc}}$ is the effective electric field acting on the Cr ions and N is the number of chromium ions per unit volume.

The $\mu\nu$ matrix element of the polarizability tensor α is given by

$$(\alpha)_{\mu\nu} = \frac{e^2}{\hbar} \sum_e \sum_i \left(\frac{\langle g|x_i|e_i\rangle \langle e_i|x_\mu|g\rangle}{\omega_{e,g} + \omega - i\Gamma_e} + \frac{\langle g|x_\nu|e_i\rangle \langle e_i|x_\mu|g\rangle}{\omega_{e,g} - \omega + i\Gamma_e} \right) \quad (\text{A9})$$

with $\hbar\omega_{e,g} = \epsilon_{e_i} - \epsilon_g$, where $|g\rangle$ and $|e_i\rangle$ are eigenfunctions of the $S_z = 3/2$ component of the ground state and of the i -th component of one of the excited 4T_2 states. ϵ_g and ϵ_{e_i} are the energies of the respective states. In this expression

we have introduced a damping factor Γ which depends on each excited state e , by taking the effects of absorption into account.

In the case where the sample is magnetized along the c axis (z axis), the system has trigonal symmetry, and therefore, from the symmetry consideration for the tensor, we can show that α has the following form:

$$\alpha = \begin{bmatrix} \alpha_{\perp} & \alpha' & 0 \\ -\alpha' & \alpha_{\perp} & 0 \\ 0 & 0 & \alpha_{\parallel} \end{bmatrix} \quad (\text{A10})$$

For the sake of simplicity, we assume that the local field is approximately equivalent to the external electric field. In such instance, from equations (A5), (A6), (A7) and (A10) we obtain

$$(n^2 - 1 - 4\pi N\alpha_{\perp})E_x - 4\pi N\alpha'E_y = 0, \\ 4\pi N\alpha'E_x + (n^2 - 1 - 4\pi N\alpha_{\perp})E_y = 0.$$

In order to get nontrivial solutions for E_x and E_y we must have

$$n^2 = 1 + 4\pi N(\alpha_{\perp} \pm i\alpha'). \quad (\text{A11})$$

For the solution of $E_y = iE_x$,

$$n_+^2 = 1 + 4\pi N(\alpha_{\perp} + i\alpha'), \quad (\text{A12})$$

while for $E_y = -iE_x$,

$$n_-^2 = 1 + 4\pi N(\alpha_{\perp} - i\alpha'). \quad (\text{A13})$$

The complex specific rotation $\rho^c(\omega)$ is defined by

$$\rho^c(\omega) = \frac{\omega}{2c}(n_+ - n_-) \quad (\text{A14})$$

Using (A12), (A13) and the mean index of refraction $n = (n_+ + n_-)/2$,

$$\rho^c(\omega) = \frac{2\pi i\omega}{nc} N\alpha' \quad (\text{A15})$$

By using eigenfunctions (3a) and (3c) of Part II, Section 3, we have computed the matrix element of dipole moment

$$\langle m = \frac{1}{2} | x | g \rangle = \frac{a}{\sqrt{2}}$$

$$\langle m = \frac{1}{2} | y | g \rangle = i \frac{a}{\sqrt{2}}$$

$$\langle m = \frac{5}{2} | x | g \rangle = -\frac{1}{\sqrt{2}}$$

$$\langle m = \frac{5}{2} | y | g \rangle = \frac{i}{\sqrt{2}}$$

where the common factor has been dropped, and a is the coefficient of component $|L_z = \mp 1, S_z = \pm \frac{3}{2}\rangle$ in eigenfunction (3a).

From (A9) and (A10)

$$\alpha' = \frac{i}{\hbar} p^2 \left\{ \frac{\omega - i\Gamma}{(\omega_{cg}^2 - \omega^2 + \Gamma^2) + 2i\omega\Gamma} - \sum_{i=1,2,3} \frac{(\omega - i\Gamma)\alpha_{A_i}^2}{(\omega_{A_i g}^2 - \omega^2 + \Gamma^2) + 2i\omega\Gamma} \right\} \quad (\text{A16})$$

where $\hbar\omega_{cg} = \epsilon_c - \epsilon_g$, $\hbar\omega_{Ag} = \epsilon_A - \epsilon_g$ and p^2 is the average dipole moment for the transition from the ground state to excited $4T_2$ state. In equation (A16) we have considered only one excited $4T_2$ state. Since in chromium trihalides more than one electron configuration has T_2 symmetry, actually we have to sum the expression of (A16) over the all electron configurations having T_2 symmetry. The expression (A16) has an essentially equivalent form to that derived by Shen for the Faraday rotation of rare-earth ions.⁽²⁶⁾

If we take $S_z = -3/2$ as a lowest component of the ground state for reversing the direction of the magnetization, it is easily shown that an x -component of the electric dipole moment changes its sign while a y -component does not for the reversal of the magnetization. Thus, we have Onsager's relation:

$$\alpha_{xy}(M) = \alpha_{yx}(-M), \quad (\text{A17})$$

where $\alpha' = \alpha_{xy}(M)$.

By substituting (A16) into (A15) we obtain the dispersion formula for the magnetic rotation $\rho^c(\omega)$. Since we have considered the effect of absorption, the refracted beam becomes elliptically polarized. Experimentally, however, we could not detect the ellipticity. Therefore, here we omit an expression for the ellipticity.

APPENDIX III THE SPIN-ORBIT COUPLING CONSTANT IN THE ELECTRON TRANSFER STATES

As was shown by TANABE and KAMIMURA,⁽²⁷⁾ the matrix element of a component ($q, \bar{\gamma}$) of the spin-orbit interaction V_{so} between two states $2^{S+1}\Gamma$ and $2^{S'+1}\Gamma'$ can be factored as follows:

$$\begin{aligned} & (S\Gamma M_S \gamma | V_{so}(q, \bar{\gamma}) | S'\Gamma' M_S' \gamma') \\ &= \frac{1}{\sqrt{[\Gamma(2S+1)]}} (S\Gamma || V_{so} || S'\Gamma') \\ & \quad \times (\Gamma \gamma | \Gamma' \gamma' T_1 \bar{\gamma}) (S M_S | S' M_S' 1 q) \end{aligned} \quad (B1)$$

where $(\Gamma \gamma | \Gamma' \gamma' T_1 \bar{\gamma})$ and $(S M_S | S' M_S' 1 q)$ are the Clebsch-Gordon coefficients and the reduced matrix $(S\Gamma || V_{so} || S'\Gamma')$ is independent of $\gamma, \gamma', \bar{\gamma}$ and M_S, M_S', q . We have specified a component of the state $2^{S+1}\Gamma$ by a wave function $\Psi(S, \Gamma, M_S \gamma)$, where $M_S(-S, -S+1, \dots, S)$ is a z component of the spin angular momentum S and γ is a component of Γ degenerate orbital states.

On the other hand, in the effective Hamiltonian (2) we have expressed the spin-orbit interaction V_{so} for the $4T_2$ state in the form $-\lambda(\alpha^4 T_2) L \cdot S$. Therefore, we have

$$\begin{aligned} & (S\Gamma M_S \gamma | V_{so}(q, \bar{\gamma}) | S'\Gamma' M_S' \gamma') \\ &= -\lambda (\Gamma \gamma S M_S | L \bar{\gamma} | \Gamma' \gamma' S M_S') (\Gamma \gamma S M_S | S_q | \Gamma \gamma S M_S') \\ &= -\lambda \frac{1}{\sqrt{[\Gamma(2S+1)]}} (\Gamma || L || \Gamma') (S || S || S) (\Gamma \gamma | \Gamma' \gamma' T_1 \bar{\gamma}) \\ & \quad \times (S M_S | S' M_S' 1 q) \end{aligned} \quad (B2)$$

In (B1) and (B2), we have used the fact that the orbital and spin angular momenta L and S are the operators of the types T_{1g} and $S = 1$, respectively.

From (B1) and (B2), the spin-orbit coupling constant for the state $2^{S+1}\Gamma$ is defined by

$$\lambda(\alpha^{2S+1}\Gamma) = -(S\Gamma || V_{so} || S\Gamma) / (S || S || S) (\Gamma || L || \Gamma) \quad (B3)$$

It can be shown that

$$\begin{aligned} (S || S || S) &= \sqrt{[S(S+1)(2S+1)]}^{(28)} \\ (T_2 || L || T_2) &= -\sqrt{(6)} i^{(27)} \end{aligned} \quad (B4)$$

Thus it is necessary to calculate only the reduced matrix of the spin-orbit interaction V_{so} to obtain λ . In the present case, the electron transfer state has the electron configuration $t_{2g}^3 e_g^* t_{1u}^5$ (or $t_{2g}^3 e_g^* t_{2u}^5$). In this configuration the three t_{2g}^* electrons have parallel spins, and couple with the e_g^* electron, also with parallel spin to form a 5E_g state. These electrons couple with the five t_{1u} electrons, two with parallel, three with antiparallel spins, so that the total state of the whole configuration is 4T_2 . We specify this configuration by

$$\Psi(t_{2g}^3(^4A_{2g})e_g(^2E_g)(^5E_g)t_{1u}^5(^2T_{1u})^4T_{2u}) \quad (B5)$$

neglecting asterisks. In this configuration we have neglected the 3E_g intermediate state in which the e_g^* electron has an antiparallel spin to the three t_{2g}^* electrons, because the 3E_g intermediate state has higher energy than the 5E_g state. However, even though we take into account the 3E_g state as well as the 5E_g state, we can show that the final formulae for the spin-orbit constants are essentially the same except numerical factors. Thus it is sufficient in the present treatment to consider only the 5E_g intermediate state.

By equation (B1) we must compute

$$\begin{aligned} & (t_{2g}^3(^4A_{2g})e_g(^2E_g)(^5E_g)t_{1u}^5(^2T_{1u})^4T_{2\gamma} M_S | V_{so}(1q T_1 \bar{\gamma}) | t_{2g}^3(^4A_{2g})e_g(^2E_g)(^5E_g)t_{1u}^5(^2T_{1u})^4T_{2\gamma'} M_S') \\ &= \frac{1}{\sqrt{3 \cdot 4}} (t_{2g}^3(^4A_{2g})e_g(^2E_g)(^5E_g)t_{1u}^5(^2T_{1u})^4T_{2u} || V(1 T_1) || t_{2g}^3(^4A_{2g})e_g(^2E_g)(^5E_g)t_{1u}^5(^2T_{1u})^4T_{2u}) \\ & \quad \times (T_{2\gamma} | T_{2\gamma'} T_1 \bar{\gamma}) (\frac{3}{2} M_S | \frac{3}{2} M_S' 1 q). \end{aligned} \quad (B6)$$

As was mentioned in (B5), the wave function Ψ is formed by coupling a normalized antisymmetrized state

$$\phi(t_{2g}^3(4A_{2g})e_g(2E_g)^5E_g\delta m_s)$$

to a normalized antisymmetrized state

$$\phi(t_{1u}^5{}^2T_{1u}\delta' m_s') \text{ (or } \phi(t_{2u}^5{}^2T_{2u}\delta' m_s')).$$

Thus

$$\begin{aligned} & \Psi(^4T_{2u}M_s\gamma) \\ &= \sum_{\substack{mm_s' \\ \delta\delta'}} \phi(t_{2g}^3(4A_{2g})e_g(2E_g)^5E_g\delta m_s) \\ & \quad \times \phi(t_{1u}^5{}^2T_{1u}\delta' m_s') \\ & \quad \times (2m_s\frac{1}{2}m_s'|\frac{3}{2}M_s)(E_g\delta T_{1u}\delta'|T_{2u}\gamma) \end{aligned} \quad (B7)$$

where Ψ is antisymmetrized and normalized.

We assume that the spin-orbit interaction V_{so} is expressed by the sum of the spin-orbit interaction of the single electron v_{so} as follows;

$$V_{so} = \sum_{i=1}^9 v_{so}(i). \quad (B8)$$

Thus, we have

$$\begin{aligned} (^4T_{2\gamma}M_s|V_{so}|^4T_{2\gamma'}M_s') &= \sum_{\substack{m_s'm_s'm \\ \delta'\delta'\delta}} (t_{1u}^5{}^2T_{1u}\delta' m_s'|V_{so}|t_{1u}^5{}^2T_{1u}\delta' m_s')(T_{2u}\gamma|T_{1u}\delta'E_g\delta) \\ & \quad \times (T_{1u}\delta'E_g\delta|T_{2u}\gamma')(\frac{3}{2}M_s|\frac{1}{2}m_s'2m_s)(\frac{1}{2}m_s'm_s|\frac{3}{2}M_s') \\ & + \sum_{\substack{m_s'm_s'm \\ \delta\delta\delta'}} (t_{2g}^3(4A_{2g})e_g(2E_g)^5E_g\delta m_s|V_{so}|t_{2g}^3(4A_{2g})e_g(2E_g)^5E_g\delta m_s) \\ & \quad \times (T_{2u}\gamma|T_{1u}\delta'E_g\delta)(T_{1u}\delta'E_g\delta|T_{2u}\gamma')(\frac{3}{2}M_s|\frac{1}{2}m_s'2m_s)(\frac{1}{2}m_s'2m_s|\frac{3}{2}M_s'). \end{aligned} \quad (B9)$$

Since

$$(t_{2g}^3(4A_{2g})e_g(2E_g)^5E_g\delta m_s|V_{so}|t_{2g}^3(4A_{2g})e_g(2E_g)^5E_g\delta m_s) = 0,$$

the second term of (B9) vanishes. Thus

$$\begin{aligned} & (^4T_{2\gamma}M_s|V_{so}|^4T_{2\gamma'}M_s') \\ &= \sum_{\substack{m_s'm_s'm_s \\ \delta'\delta'\delta}} \frac{1}{\sqrt{2\cdot 3}} (t_{1u}^5{}^2T_{1u}|V_{so}|t_{1u}^5{}^2T_{1u}) \\ & \quad \times (T_{1u}\delta'|T_{1u}\delta'T_{1g}\bar{\gamma}) \\ & \quad \times (T_{2u}\gamma|T_{1u}\delta'E_g\delta)(T_{1u}\delta'E_g\delta|T_{2u}\gamma')(\frac{1}{2}m_s'|\frac{1}{2}m_s'1q) \\ & \quad \times (\frac{3}{2}M_s|\frac{1}{2}m_s'2m_s)(\frac{1}{2}m_s'2m_s|\frac{3}{2}M_s'). \end{aligned} \quad (B10)$$

Using the orthonormal relation of Clebsch-Gordon coefficients the following relation is derived from (B6) and (B10):

$$\begin{aligned} (^4T_2|V_{so}|^4T_2) &= \sqrt{(2)}(t_{1u}^5{}^2T_{1u}|V_{so}|t_{1u}^5{}^2T_{1u}) \\ & \quad \times (T_{1u}T_{1g}(T_{1u})E_gT_{2u}|T_{1g},T_{1u}E_g(T_{2u})T_{2u}) \\ & \quad \times (\frac{1}{2}1(\frac{1}{2})2\frac{3}{2}|1,\frac{1}{2}2(\frac{3}{2})\frac{3}{2}) \end{aligned} \quad (B11)$$

where we have used the Racah's transformation matrices defined in the following:⁽²⁹⁾

$$\begin{aligned} & (T_{1u}T_{1g}(T_{1u})E_gT_{2u}|T_{1g},T_{1u}E_g(T_{2u})T_{2u}) \\ &= \sum_{\delta',\delta',\delta',\gamma',\gamma} (T_{1u}\delta'|T_{1u}\delta'T_{1g}\bar{\gamma})(T_{2u}\gamma|T_{1u}\delta'E_g\delta) \\ & \quad \times (T_{1u}\delta'E_g\delta|T_{2u}\delta')(T_{2u}\gamma'T_{1g}\bar{\gamma}|T_{2u}\gamma) \\ & \quad (\frac{1}{2}1(\frac{1}{2})2\frac{3}{2}|1,\frac{1}{2}2(\frac{3}{2})\frac{3}{2}) \\ &= \sum_{m_s',m_s',m_s,q,M_s} (\frac{1}{2}m_s'|\frac{1}{2}m_s'1q)(\frac{3}{2}M_s|\frac{1}{2}m_s'2m_s) \\ & \quad (\frac{1}{2}m_s'2m_s|\frac{3}{2}M_s')(\frac{3}{2}M_s'1q|\frac{3}{2}M_s). \end{aligned} \quad (B13)$$

We have computed

$$(T_{1u}T_{1g}(T_{1u})E_gT_{2u}|T_{1g},T_{1u}E_g(T_{2u})T_{2u}) = \frac{1}{2}$$

and

$$(\frac{1}{2}1(\frac{1}{2})2\frac{3}{2}|1,\frac{1}{2}2(\frac{3}{2})\frac{3}{2}) = \frac{4}{\sqrt{5}}.$$

Since

$$(t_{1u}^5{}^2T_{1u}|V_{so}|t_{1u}^5{}^2T_{1u}) = -(t_{1u}^2T_{1u}|V_{so}|t_{1u}^2T_{1u}),$$

we obtain from (B3)

$$\lambda(^4T_2(t_{2g}^3e_g t_{1u}^5)) = -\frac{2}{15}i(\frac{1}{2}t_{1u}|v_{so}|\frac{1}{2}t_{1u}) \quad (B14)$$

where v_{so} is the single electron operator of the spin-orbit interaction.

Similarly,

$$\lambda(4T_2(t_{2g}^3 e_g t_{2u}^5)) = -\frac{2}{15}i(\frac{1}{2}t_{2u}||v_{so}||\frac{1}{2}t_{2u}). \quad (B15)$$

Nonbonding t_{1u}^n and t_{2u}^n orbitals and bonding t_{1u} molecular orbitals are shown in Table B1. In each ket vector of the table the first letter denotes the type of orbit, the second letter the angular dependence, and the number the particular halogen ion on which it is centered. As shown in Fig. 6, the halogen ions 1, 2 and 3 lie on the positive X , Y and Z axes, and 4, 5 and 6 on the negative X , Y and Z axes. The symbols ϕ_{4p} 's are the $4p$ wave functions of a Cr^{3+} ion, μ gives a measure of the amount of halogen p orbitals, and N is a normalizing factor. Since $(\frac{1}{2}t_{1u}\frac{1}{2}\alpha|\zeta 1_z S_z|\frac{1}{2}t_{1u}\frac{1}{2}\beta) = -\frac{1}{2}(\frac{1}{2}t_{1u}||v_{so}||\frac{1}{2}t_{1u})$, for the nonbonding t_{1u}^n orbitals $(\frac{1}{2}t_{1u}||v_{so}||\frac{1}{2}t_{1u}) = \frac{3}{2}i\zeta_x$. Here ζ_x is the spin-orbit coupling constant of the single p electron of the halogen atom X . Similarly, for the nonbonding t_{2u}^n orbitals we obtain

$$(\frac{1}{2}t_{2u}||v_{so}||\frac{1}{2}t_{2u}) = -\frac{3}{2}i\zeta_x.$$

Thus for the electron transfer states from a non-bonding t_{1u}^n (or t_{2u}^n) to an e_g state

$$\lambda(4T_2) = -\frac{1}{5}\zeta_x. \quad (B16)$$

For the bonding t_{1u} orbital which has mainly halogen character ($\mu \gg 1$), $(\frac{1}{2}t_{1u}\frac{1}{2}\alpha|\zeta 1_z S_z|\frac{1}{2}t_{1u}\frac{1}{2}\beta') = 0$, as far as we neglect the overlap between p_σ orbitals on the adjacent halogen ions.

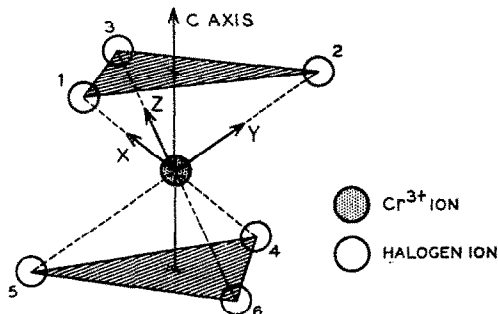


FIG. 6. Our model for CrBr_3 . The Cr ions are between equally spaced planes of Br ions. Each Cr ion is in octahedral bromine coordination, but the octahedron is severely distorted along the c axis with the point symmetry at the Cr sites being C_3 .

Taking the effect of this overlap into account, we have approximately

$$(\frac{1}{2}t_{1u}\frac{1}{2}\alpha|\zeta 1_z S_z|\frac{1}{2}t_{1u}\frac{1}{2}\beta') \approx -\frac{i\mu^2}{2N}S_x\zeta_x,$$

Table B1. The t_{1u} molecular orbital and t_{1u}^n and t_{2u}^n nonbonding orbitals

t_{1u}^n	$ \alpha\rangle = \frac{1}{2}\{ p, X, 2\rangle + p, X, 5\rangle + p, X, 3\rangle + p, X, 6\rangle\}$
	$ \beta\rangle = \frac{1}{2}\{ p, Y, 1\rangle + p, Y, 4\rangle + p, Y, 3\rangle + p, Y, 6\rangle\}$
	$ \gamma\rangle = \frac{1}{2}\{ p, Z, 1\rangle + p, Z, 4\rangle + p, Z, 2\rangle + p, Z, 5\rangle\}$
t_{2u}^n	$ \xi\rangle = \frac{1}{2}\{ p, X, 2\rangle + p, X, 5\rangle - p, X, 3\rangle - p, X, 6\rangle\}$
	$ \eta\rangle = \frac{1}{2}\{- p, Y, 1\rangle - p, Y, 4\rangle + p, Y, 3\rangle + p, Y, 6\rangle\}$
	$ \zeta\rangle = \frac{1}{2}\{ p, Z, 1\rangle + p, Z, 4\rangle - p, Z, 2\rangle - p, Z, 5\rangle\}$
t_{1u}	$ \alpha'\rangle = \frac{1}{\sqrt{N}}\left\{\phi_{4pX} + \frac{\mu}{\sqrt{2}}(p, X, 1\rangle + p, X, 4\rangle)\right\}$
	$ \beta'\rangle = \frac{1}{\sqrt{N}}\left\{\phi_{4pY} + \frac{\mu}{\sqrt{2}}(p, Y, 2\rangle + p, Y, 5\rangle)\right\}$
	$ \gamma'\rangle = \frac{1}{\sqrt{N}}\left\{\phi_{4pZ} + \frac{\mu}{\sqrt{2}}(p, Z, 3\rangle + p, Z, 6\rangle)\right\}$

where

$$S_x = 2 \langle p, Y, 2 | p, Y, 1 \rangle.$$

Thus, for the electron transfer state from a bonding molecular orbital t_{1u} to an e_g^* state,

$$\lambda(^4T_2) = -\frac{2}{5} \frac{\mu^2}{N} S_x \zeta_x. \quad (\text{B17})$$

On the other hand, in the case of the intra-atomic type transition for which the final state has the electron configuration $t_{2g}^{*2}t_{1u}^*$, the spin-orbit coupling constant $\lambda(^4T_2(t_{2g}^{*2}t_{1u}^*))$ is

calculated similarly. In this case,

$$\begin{aligned} & (t_{2g}^2(^3T_1)t_{1u}(^2T_{1u})^4T_2 \| V_{so} \| t_{2g}^2(^3T_1)t_{1u}(^2T_{1u})^4T_2) \\ &= \{ (t_{2g}^2(^3T_{1u}) \| V_{so} \| t_{2g}^2(^3T_{1g}) + (t_{1u}^2T_{1u} \| V_{so} \| t_{1u}^2T_{1u}) \} \\ & \quad \times \sqrt{\left(\frac{4}{3}\right)} \left(11\left(1\right)\frac{1}{2}\frac{3}{2} \middle| 1, 1\frac{1}{2}\left(\frac{3}{2}\right)\frac{3}{2}\right) \\ & \quad \times \langle T_{1g}T_{1u}(T_{2u})T_{1g}T_{2u} | T_{1u}, T_{1g}T_{1g}(T_{1g})T_{2u} \rangle \\ &= \frac{3\sqrt{10}}{2} i(\zeta_{4p} - \zeta_{3d}) \end{aligned} \quad (\text{B18})$$

where ζ_{3d} and ζ_{4p} are the spin-orbit coupling constants of the single $3d$ and $4p$ electrons of a Cr^{3+} ion, respectively. Thus, from (B3),

$$\lambda(^4T_2(t_{2g}^2t_{1u})) = \frac{1}{2}(\zeta_{4p} - \zeta_{3d}). \quad (\text{B19})$$